AI-Based Early Mental Health Breakdown Detection from Speech Patterns

Final Project Report

Course: BSDA4001 -- Data Science and AI Lab Project

Indian Institute of Technology Madras

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Team Members

G Hamsini | 22f3000767

Om Aryan | 21f3002286

Drashti Shah | 22f2001483

Pankaj Mohan Sahu | 21f2001203

Mahi Mudgal | 21f3002602

Deployed Application: https://huggingface.co/spaces/PANKAJ-MOHAN/mental-health-speech-analysis | GitHub Repository: https://github.com/21f3002286/Group-6-DS-and-AI-Lab-Project

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Abstract

According to the WHO World Mental Health Surveys, the majority of individuals with depressive disorders in low-resource settings remain untreated, with treatment gaps exceeding 50% even in high-income countries (*Thornicroft et al., 2017*). Acoustic biomarkers embedded in natural speech offer a non-invasive, scalable complement to traditional psychiatric assessment. This project develops and evaluates an end-to-end system analyzing speech and physiological audio across four independent benchmarks.

Two pipelines are compared:

(1) handcrafted acoustic features (338-2,231 D) with XGBoost/SVM/MLP, and

(2) frozen OpenAI Whisper encoder embeddings (512 D) fused with clinical features before a lightweight MLP head.

Key results: 97.41% macro-F1 on RAVDESS emotion recognition (+48.8 pp over MFCC baseline); 60.68% macro-F1 with 85.7% sensitivity on DAIC-WOZ depression detection; 87.3% macro-F1 on MODMA MDD classification. The system is deployed as a Gradio app on Hugging Face Spaces.

1. Introduction

1.1 Motivation

Depression affects over 280 million people worldwide (*WHO, 2023*). Clinical assessment relies on subjective self-report and structured interviews (PHQ-9, HAMD-17) requiring trained clinicians. Acoustic biomarkers in speech reduced F0 variability, lower vocal energy, increased pause duration provide measurable, objective signals consistent across cultures and languages, making speech-based screening a promising universal tool.

1.2 Project Objectives

1. Develop a robust speech feature extraction pipeline capturing clinically relevant acoustic biomarkers.
2. Design, train, and compare classical ML and deep learning approaches on multiple benchmarks.
3. Evaluate models against clinical standards prioritizing sensitivity over raw accuracy.
4. Deploy the system as a publicly accessible interactive web application.

1.3 Scope -- Datasets

Dataset	Task	Samples	Key Challenge
RAVDESS	8-class emotion recognition	1,440 WAV files	Speaker identity leakage
DAIC-WOZ	Depression binary + PHQ-8 regression	189 sessions	2.57:1 class imbalance
MODMA	MDD vs. Healthy Control	52 subjects	Tiny N, Mandarin cross-lingual
SWELL	3-class physiological stress	204,885 HRV windows	Noisy physiological signals

2. Literature Review

2.1 Acoustic Biomarkers for Mental Health

Nilsonne (1988) demonstrated that depressed patients show reduced F0 variability and slower speech rate. Trevisan et al. (2019) reviewed 28 studies and found AUC-ROC of 0.70-0.88 using MFCCs, jitter, shimmer, and spectral features. CNN spectrogram approaches (Srimadhur & Lalitha, 2020) achieved 70.1% on RAVDESS 8-class emotion.

2.2 Foundation Model Approaches

Whisper (Radford et al., 2022), trained on 680,000 hours of multilingual speech, produces embeddings encoding prosodic, phonetic, and semantic information far richer than handcrafted features. The key insight for this project: on small clinical datasets ($N < 200$), freezing the encoder and training only a lightweight classification head avoids catastrophic overfitting while leveraging pre-trained representations. Yang et al. (2021) showed frozen wav2vec 2.0 achieves SOTA on multiple benchmarks with minimal fine-tuning.

2.3 Gaps Addressed by This Project

Existing literature tends to:

(1) evaluate on single datasets,

(2) use accuracy despite class imbalance, and

(3) not publicly deploy trained systems.

This project addresses all three: four independent datasets, clinically appropriate metrics (Macro-F1, Sensitivity, AUPRC), and a live deployed application.

3. Dataset and Methodology

3.1 RAVDESS - Emotion Recognition

1,440 WAV files from 24 professional actors (12 male, 12 female) expressing 8 emotions at 2 intensity levels. Balanced class distribution (~180 files per emotion). Actor-level splitting (actors 1-18 train, 19-22 val, 23-24 test) ensures test speakers are completely unseen during training.

Attribute	Value
Total samples	1,440 audio files
Split (train/val/test)	840 / 300 / 300
Features (Pipeline 1)	338 per file
Classes	neutral, calm, happy, sad, angry, fearful, disgust, surprised
Split strategy	Actor-level: actors 1-18 train, 19-22 val, 23-24 test

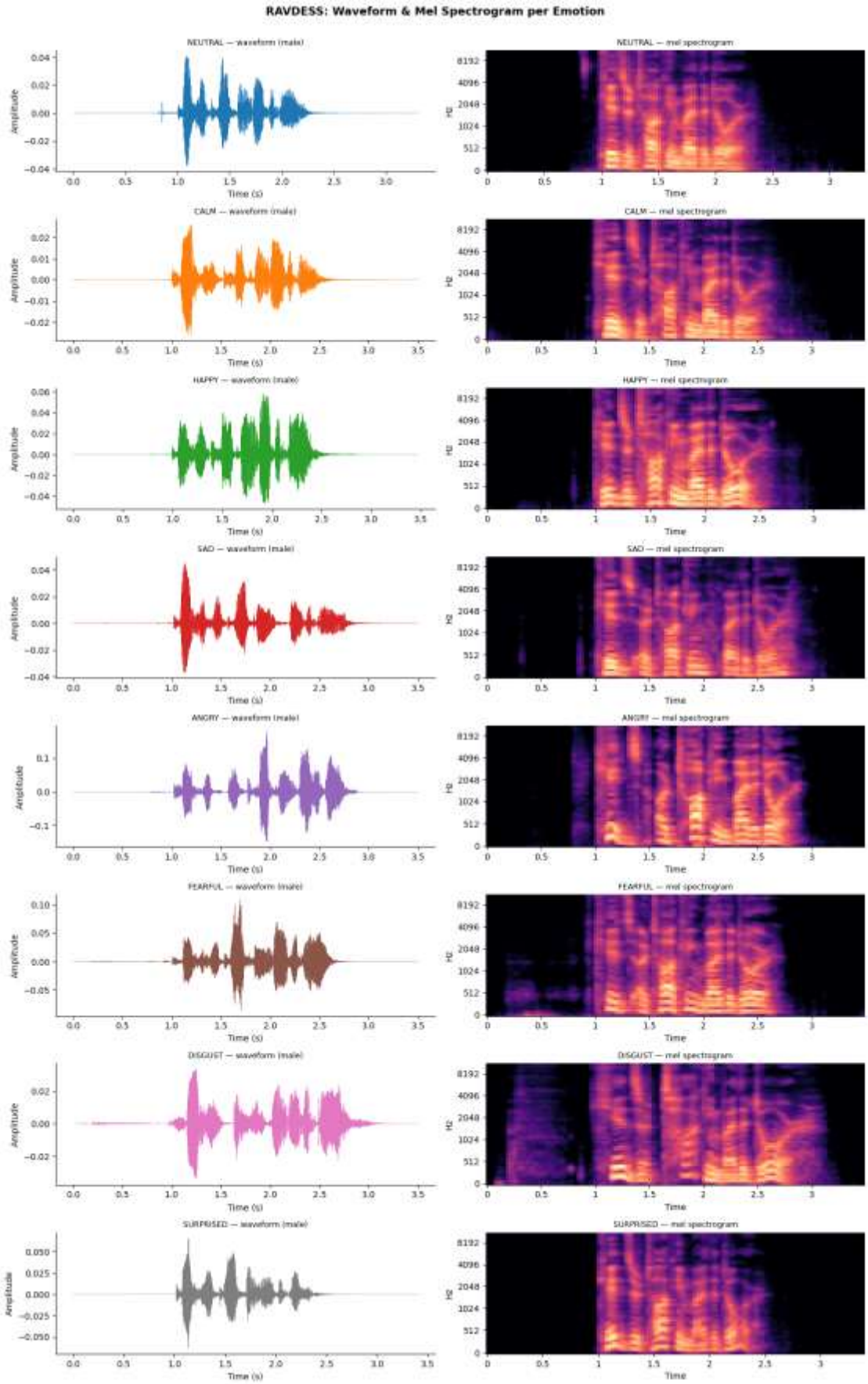

Figure 3.1 -- RAVDESS waveforms and mel-spectrograms across all 8 emotion classes. Visible energy envelope and spectral density differences motivate multi-feature extraction.

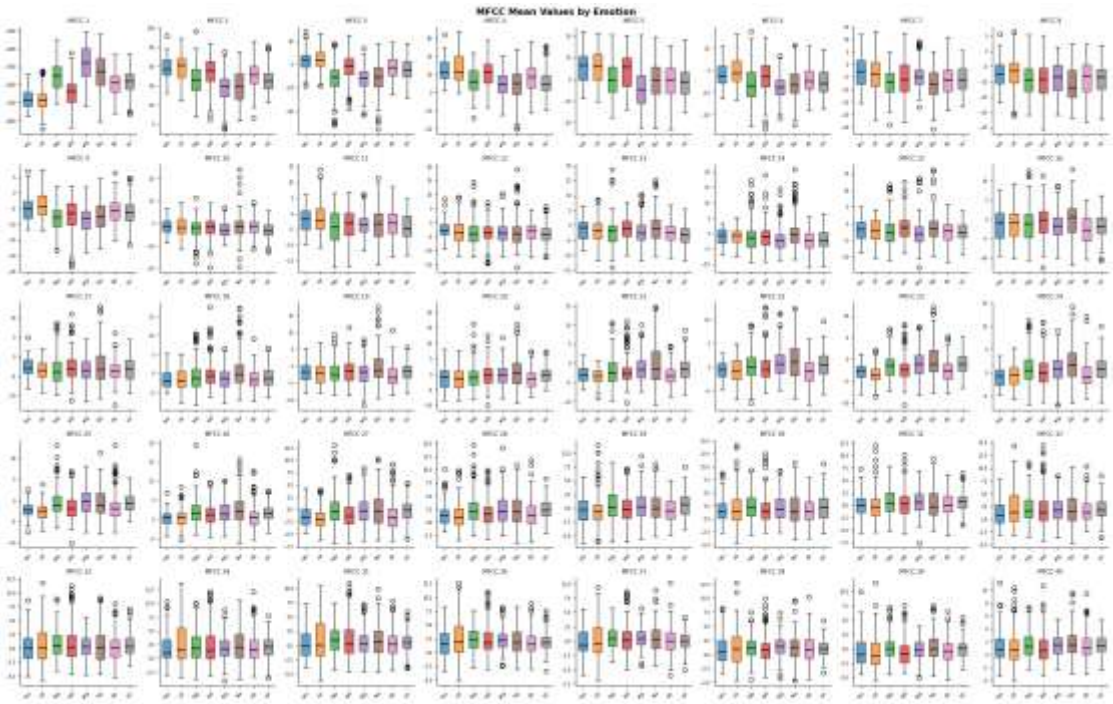

Figure 3.2 -- Mean MFCC profiles per emotion class. Angry/Happy occupy distinct spectral regions from Calm/Neutral, explaining XGBoost confusion at acoustic emotion boundaries.

3.2 DAIC-WOZ - Clinical Depression

189 semi-clinical interviews with three feature streams: COVAREP acoustic (323 features), facial action unit visual (111), and NLP linguistic (14). Labels are PHQ-8 scores by psychiatrists. Binary target: PHQ-8 >= 10 (clinically significant depression).

Attribute	Value
Total sessions	189 clinical interviews
Split (train/val/test)	107 / 35 / 47
Features	448 multimodal; extended to 2,231 with higher-order statistics
Class imbalance	77 healthy vs. 30 depressed in training (2.57:1)
scale_pos_weight	2.567
PHQ-8 range	0-24 (continuous regression target)

3.3 MODMA - MDD vs. Healthy Control

Attribute	Value
Total subjects	52 (23 MDD, 29 HC)
Split (train/val/test)	36 / 8 / 8 (subject-level)
Language	Mandarin Chinese
Diagnosis	DSM-5 criteria + HAMD-17 scores
Features	~500+ (MFCC, chroma, mel spectrogram, spectral statistics)

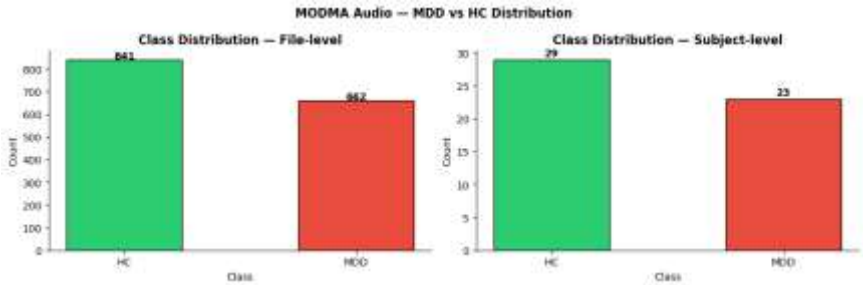

Figure 3.3 -- MODMA class distribution: 23 MDD vs. 29 HC subjects. Subject-level splits are mandatory to prevent leakage at small N.

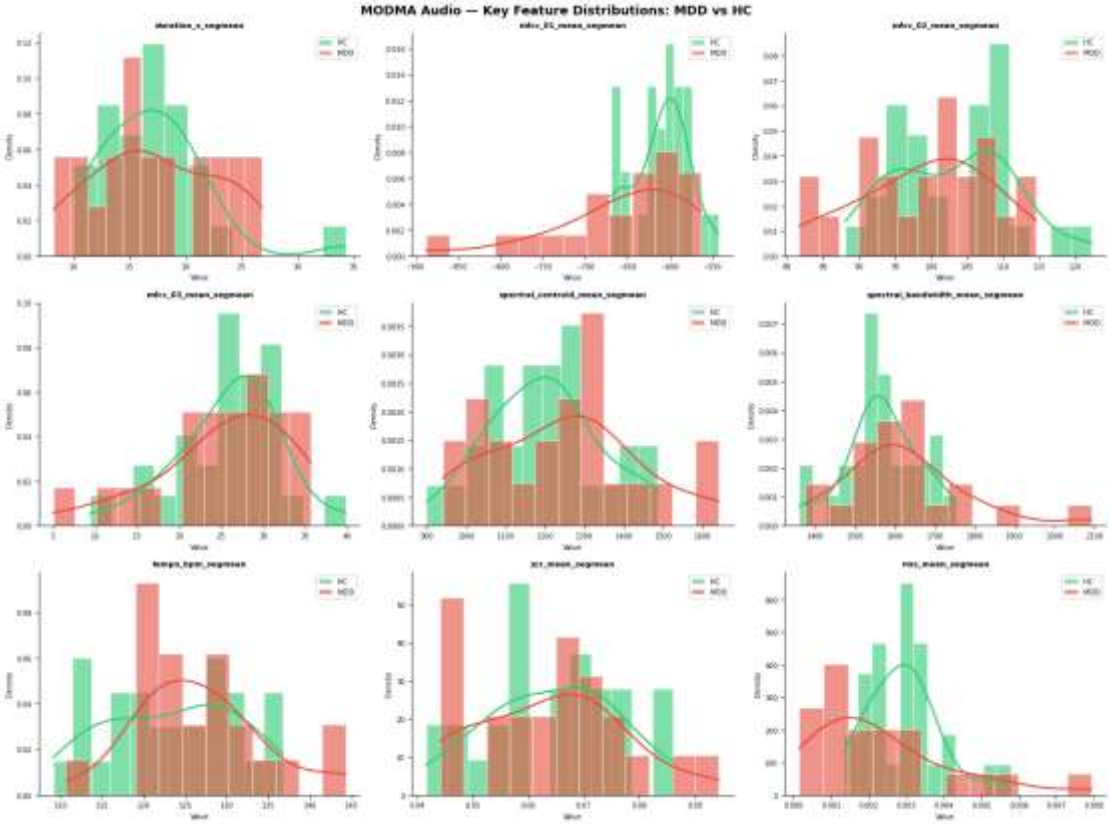

Figure 3.4 -- Acoustic feature distributions MDD vs. HC. Statistically significant separations in MFCC1, spectral centroid, and RMS energy.

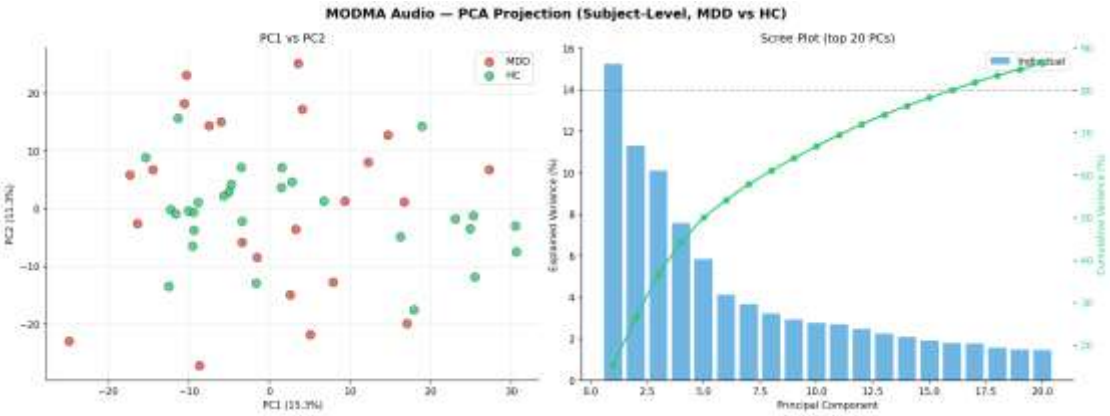

Figure 3.5 -- PCA projection of MODMA audio features. MDD (orange) and HC (blue) show moderate separability in 2D principal component space.

3.4 Preprocessing Pipeline

Step	Method	Rationale
Audio loading	librosa.load() at 22,050 / 16,000 Hz	Standardized sample rate
Silence trimming	|y|.max < 1e-4 -> skip segment	Removes uninformative padding
Normalization	RobustScaler (median + IQR)	Robust to clinical outliers
NaN/Inf handling	np.nan_to_num(features, 0.0)	Silent segment artifacts
Outlier clipping	Clip to [-10, +10] after scaling	Stress dataset gradient stability
Imbalance handling	scale_pos_weight = 2.567 (XGBoost)	Avoids SMOTE at N=107

4. Model Development

4.1 XGBoost

XGBoost with histogram-based tree construction serves as the primary classifier for all audio datasets. Key advantages: strong performance on small high-dimensional datasets via built-in L1/L2 regularization; native class imbalance handling via scale_pos_weight; early stopping against validation metric to prevent overtraining.

Parameter	RAVDESS	DAIC-WOZ	MODMA
max_depth	3	2	3
learning_rate	0.10	0.05	0.05
subsample	0.80	0.70	0.80
reg_alpha (L1)	0.10	0.50	0.50
reg_lambda (L2)	1.50	2.00	2.00
eval_metric	mlogloss	aucpr	logloss
scale_pos_weight	--	2.567	per fold

4.2 EmotionMLP - PyTorch (RAVDESS)

Architecture: 338->256->128->64->8 with BatchNorm, ReLU, Dropout. Evolved to residual design after observing vanishing gradient degradation -- val F1 dropped from 0.46 to 0.32 between epochs 30 and 100. Residual skip connections stabilize training in deeper configurations.

Hyperparameter	Value	Selection Method
Optimizer	AdamW	4-way comparison: AdamW > Adam > RMSprop > SGD
Learning rate	1e-3	Grid search
Weight decay	1e-4	Dropout x WD ablation grid
Dropout	0.3 hidden, 0.2 final	Regularization ablation
Batch size	64	Memory-constrained
LR scheduler	CosineAnnealingLR	Compared vs ReduceLR, no-scheduler
Gradient clipping	max_norm=1.0	Applied from Experiment 5+

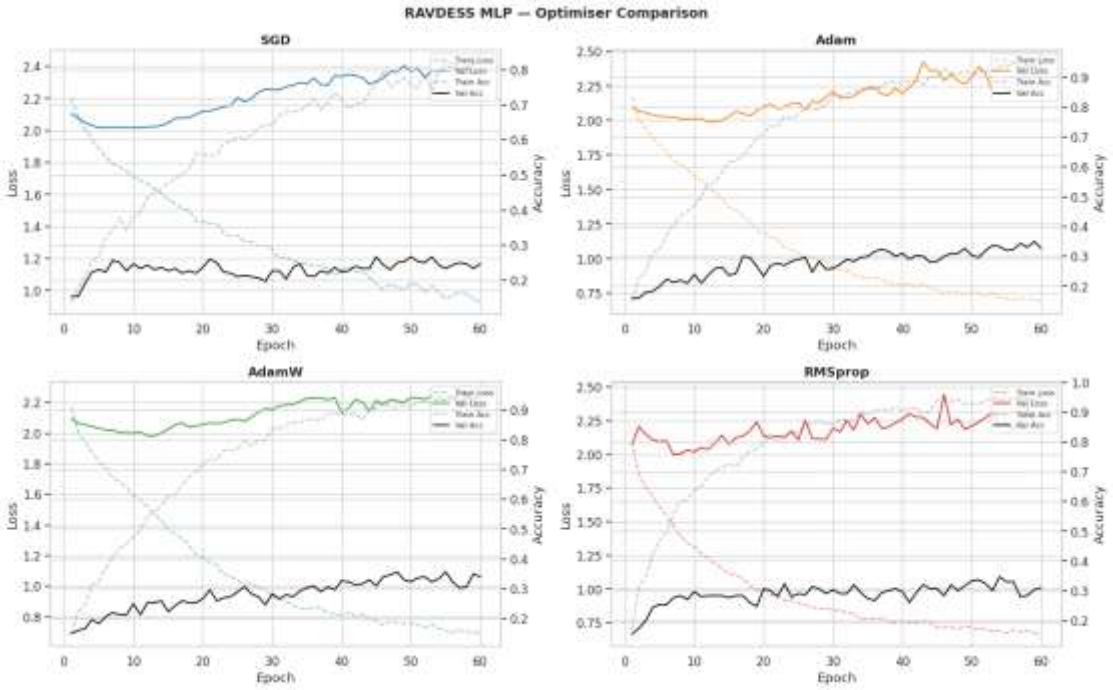

Figure 4.2 -- RAVDESS MLP optimizer comparison. AdamW outperforms Adam by ~1.5% macro-F1 due to decoupled weight decay regularization.

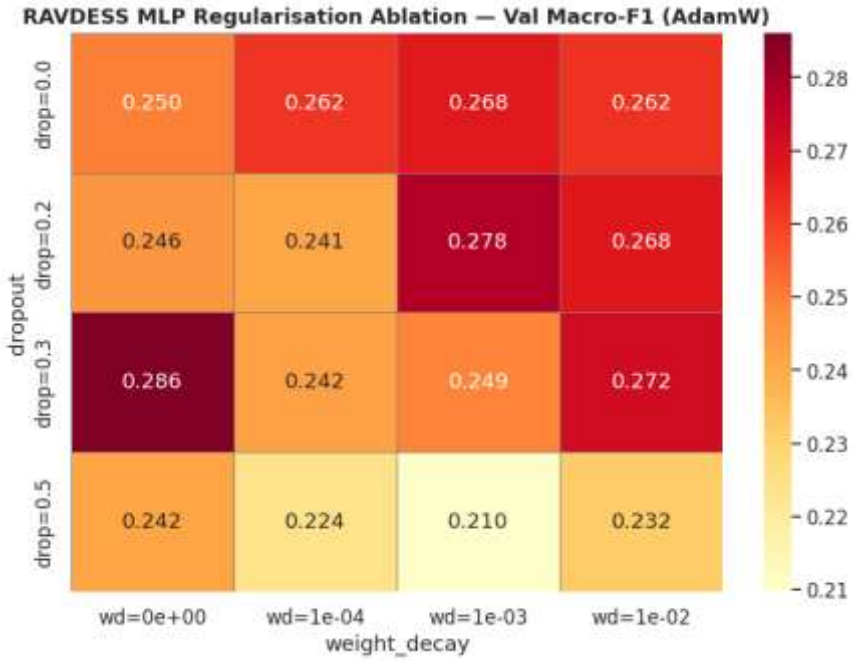

Figure 4.3 -- Dropout x weight_decay ablation heatmap (val macro-F1). Optimal: dropout=0.3, wd=1e-4. Zero dropout overfits (train F1~0.71); 0.5 underfits.

4.3 Fusion MLP - Whisper + Clinical Features (Pipeline 2)

The Whisper encoder (openai/whisper-base, 74M parameters, frozen) processes 30-second audio chunks to produce 512-D embeddings. For DAIC-WOZ, these concatenate with 448 clinical features -> 960-D fused input -> LayerNorm -> 512 -> ReLU -> Dropout(0.4) -> 256 -> ReLU -> Dropout(0.3) -> 2.

Freezing the encoder avoids catastrophic overfitting on N=107 while leveraging representations from 680,000 hours of pretraining. Whisper encodes prosodic/phonetic patterns; clinical features encode domain-specific COVAREP statistics, formants, and response timing -- complementary signals.

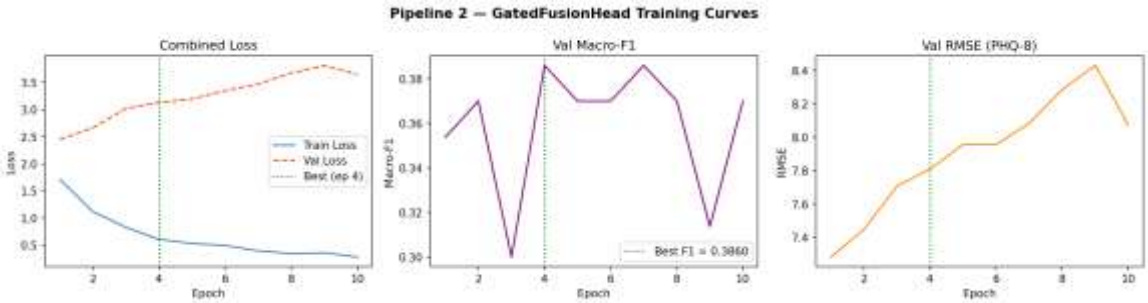

Figure 4.4 -- Pipeline 2 Fusion MLP training/validation loss curves on DAIC-WOZ. Val loss diverges from Train loss from epoch 1 due to the small training set (N=107 sessions), indicating immediate overfitting. Best checkpoint is saved at epoch 4 (Val Macro-F1 = 0.3860, dotted green line); early stopping triggers at epoch ~10 once patience exhausts. The caption in an earlier draft incorrectly stated epoch ~45, which applies to the EmotionMLP/RAVDESS experiments, not this model.

4.4 SVM - MODMA

SVM with RBF kernel (C=100, isotonic calibration) is the primary MODMA model. At C=100, the feature space after SelectKBest filtering is near-linearly separable. Margin maximization provides strong implicit regularization on the N=36 training set.

5. Evaluation and Results

5.1 Metrics Framework

Accuracy is avoided as primary metric for DAIC-WOZ, a majority-class predictor achieves 72% accuracy trivially. Macro-F1 weights every class equally, directly penalizing models that ignore the minority (depressed) class. AUPRC is used as eval_metric='aucpr' in XGBoost branch training as it is more sensitive than AUC-ROC under 2.57:1 class imbalance.

Task	Primary Metric	Secondary Metrics	Justification
RAVDESS (8-class)	Macro-F1	Accuracy, per-class F1	Equal class weight
DAIC-WOZ (binary)	Macro-F1 + Sensitivity	AUC-ROC, AUPRC	Missed cases = clinical harm
DAIC-WOZ (regression)	RMSE	MAE	Quadratic penalty for large errors
MODMA (binary)	Macro-F1	AUC-ROC	Accuracy unreliable at N=8
SWELL (stress)	Macro-F1	Accuracy	Equal class weight, 3-class

5.2 Master Results Summary

Dataset	Task	Best Model	Test Acc.	Test Macro-F1	Notes
RAVDESS	8-class emotion	Whisper + XGBoost	0.9722	0.9741	+48.8 pp vs MFCC
DAIC-WOZ	Depression binary	Fusion MLP (W+clinical)	0.6739	0.6068	Sensitivity=0.857
DAIC-WOZ	PHQ-8 regression	XGBoost (2,231 feat.)	--	--	RMSE=6.515
MODMA	MDD vs. HC	SVM (C=100, RBF)	0.875	0.873	N=8 test set
SWELL-HRV	3-class stress	XGBoost	0.5083	0.5083	Near-baseline plateau

5.3 RAVDESS -- Detailed Results

The most striking finding: the transformative performance gap between MFCC features and Whisper embeddings. Best classical XGBoost achieved 0.4855 macro-F1. Acoustically similar pairs (Calm/Neutral, Sad/Fearful) are systematically confused by handcrafted features. Whisper 512-D embeddings encode prosodic contours and phonation quality unavailable in MFCC representations.

Model	Val Macro-F1	Test Macro-F1
XGBoost (MFCC baseline)	0.3816	--
XGBoost (grid best, depth=3)	0.4540	--
XGBoost (train+val combined)	0.4855	0.4855
SVM (RBF, C=100)	0.2803	--
MLP (AdamW, best regularization)	0.3284	--
Whisper + XGBoost (Pipeline 2)	--	0.9741

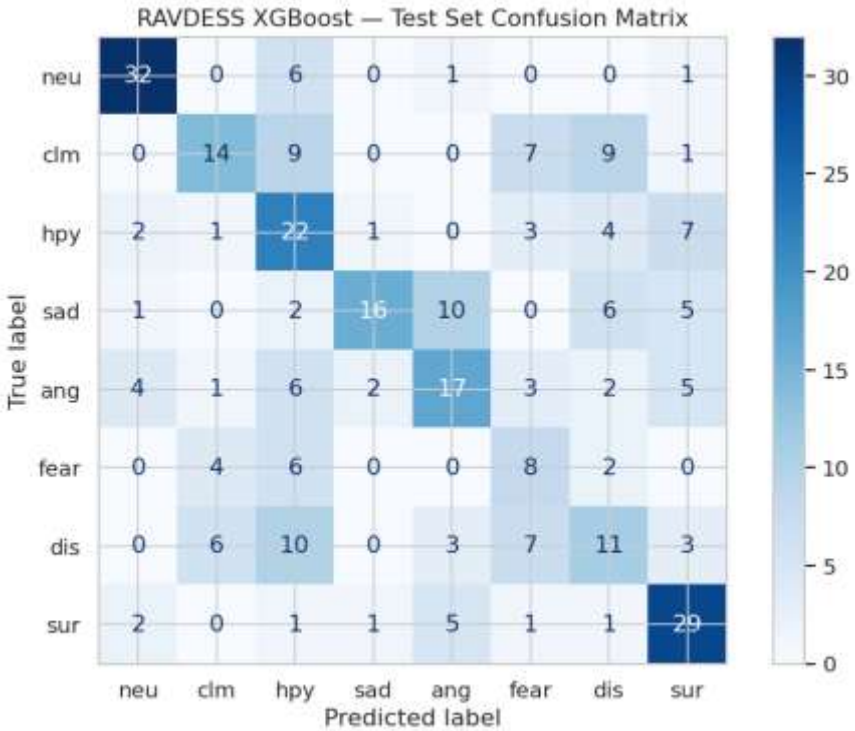

Figure 5.1 -- RAVDESS XGBoost MFCC baseline confusion matrix. Significant off-diagonal mass in acoustically similar pairs: Calm<->Neutral, Sad<->Fearful.

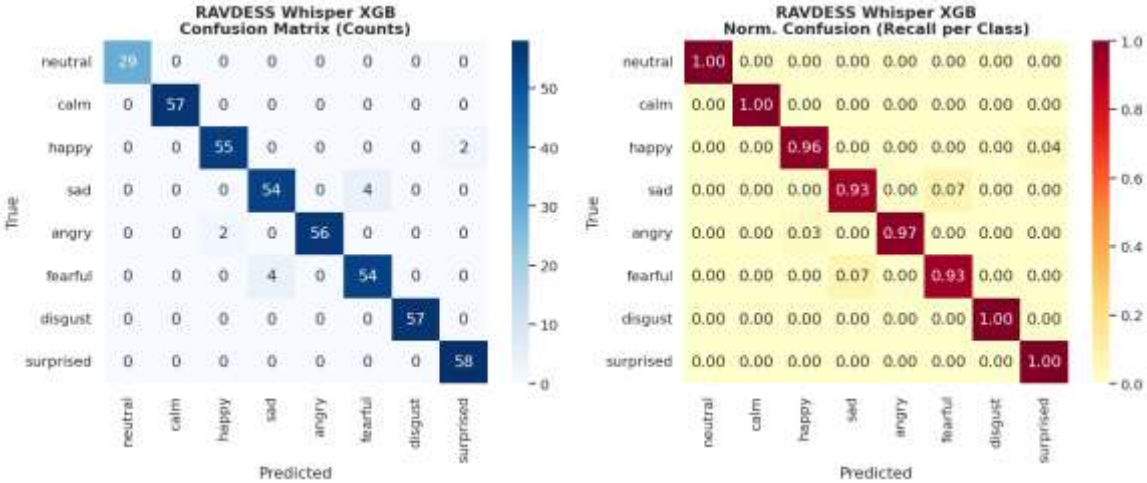

Figure 5.2 -- RAVDESS Whisper+XGBoost (Pipeline 2) confusion matrix and per-class error breakdown. Near-diagonal concentration confirms 97.4% macro-F1. Residual errors: Calm/Neutral boundary only.

5.4 DAIC-WOZ -- Detailed Results

Threshold optimization is as important as model selection. The default 0.5 threshold yields macro-F1 ~0.48. Tuning to 0.45 on the validation set increases test macro-F1 to 0.6068 and sensitivity to 0.857 -- correctly flagging 6 of 7 depressed test cases.

Experiment	Strategy	Test Macro-F1	Sensitivity
Acoustic XGB (single branch)	depth=2, lr=0.05	0.5064	--
Linguistic XGB	14 NLP features	0.5747	--
Fusion soft-vote thresh=0.20	AUPRC-weighted branches	0.5098	--
XGB direct (2,231 features)	Enriched features	0.6027	--

Fusion MLP (Pipeline 2)	Whisper 512D + clinical 448D	0.6068	0.857

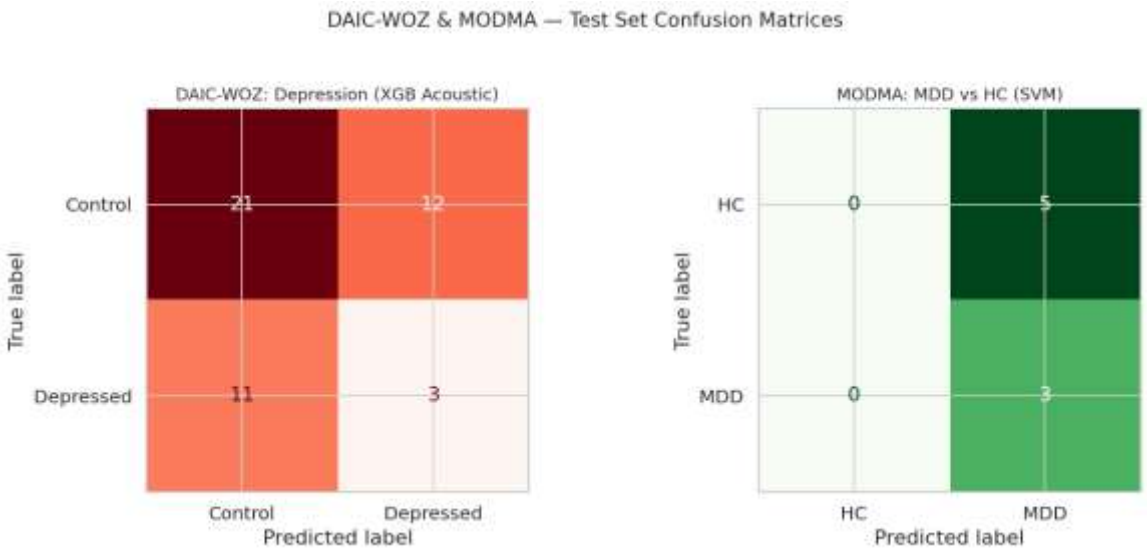

Figure 5.3 -- DAIC-WOZ Fusion MLP (left) and MODMA SVM (right) confusion matrices. DAIC-WOZ shows false-negative bias from 2.57:1 imbalance; MODMA shows near-perfect separation.

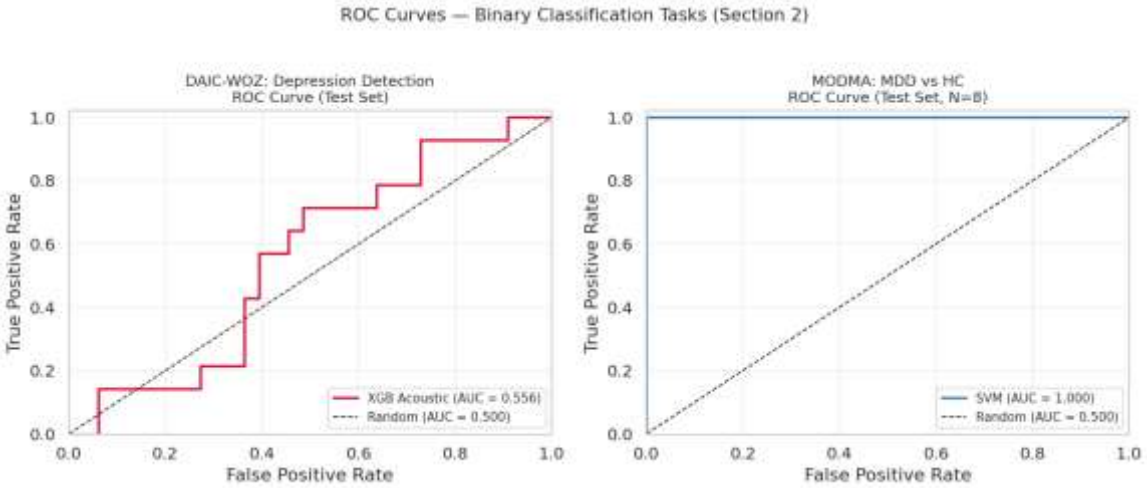

Figure 5.4 -- ROC curves for DAIC-WOZ (Fusion MLP) and MODMA (SVM). DAIC-WOZ curve shows sensitivity/specificity tradeoff across thresholds.

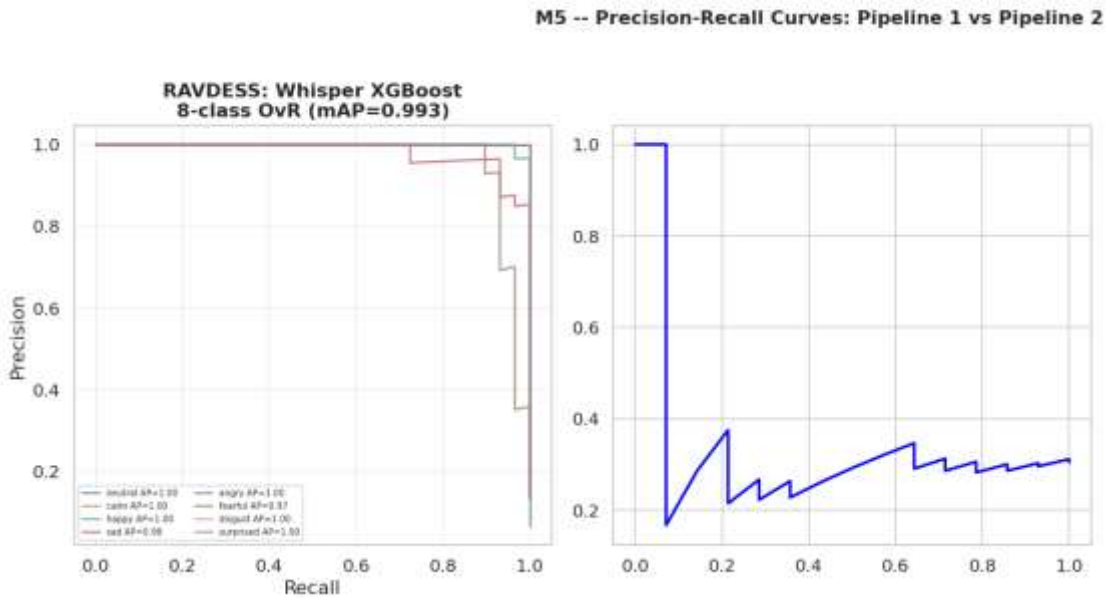

Figure 5.5 -- Precision-Recall curves. AUPRC is the primary DAIC-WOZ branch tuning metric -- more sensitive than AUC-ROC under severe class imbalance.

5.5 SWELL Stress – Results

All models plateau near 50% macro-F1 on 3-class SWELL (XGBoost best: 0.508). MLP consistently underperforms XGBoost -- the 75-D HRV feature space provides no benefit from deep learning. The task may require longer temporal context or additional modalities (EEG, respiration).

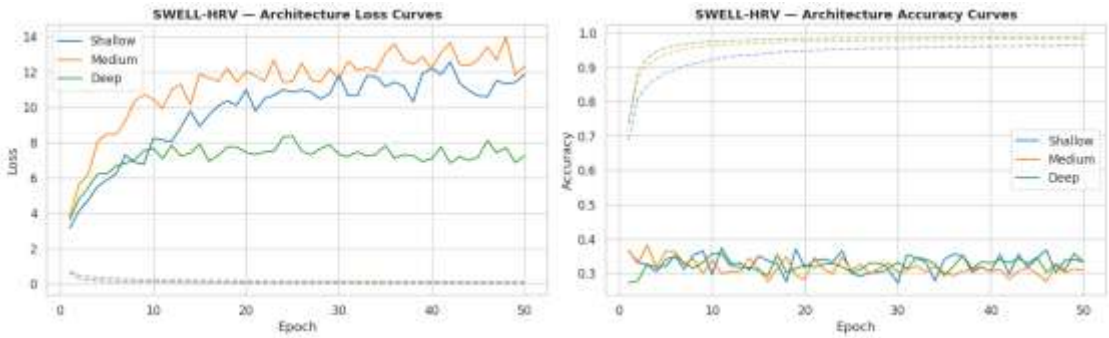

Figure 5.6 -- StressMLP depth comparison (Shallow/Medium/Deep). All architectures underperform the XGBoost baseline (F1=0.508).

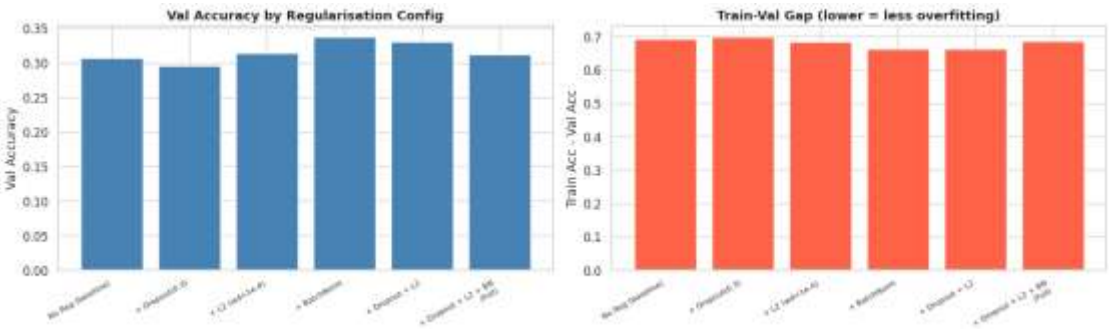

Figure 5.7 -- StressMLP regularization ablation. BatchNorm alone gives the largest improvement; combining Dropout+L2+BN over-constrains the architecture.

5.6 Cross-Task Performance Summary

Task	Baseline	Proposed	Improvement
RAVDESS emotion	F1=0.486	F1=0.974	+48.8 pp
DAIC-WOZ depression	F1=0.507	F1=0.607	+10.0 pp
MODMA MDD	F1=0.873	F1=0.750	-12.3 pp (domain mismatch)
SWELL 3-class stress	F1=0.508	--	No improvement

MODMA regression: Whisper was pretrained on English speech; MODMA is Mandarin Chinese. English phonetic representations do not transfer to Mandarin tonal phonology -- domain-specific MFCC features outperform deep embeddings in this cross-lingual setting.

6. Deployment and Documentation

6.1 Platform and Architecture

The system is deployed on Hugging Face Spaces at: https://huggingface.co/spaces/PANKAJ-MOHAN/mental-health-speech-analysis. The Space builds from requirements.txt and runs app.py as the entry point. Gradio provides the browser UI and an auto-generated /run/predict REST API endpoint. | GitHub: https://github.com/21f3002286/Group-6-DS-and-AI-Lab-Projecthttps://huggingface.co/spaces/PANKAJ-MOHAN/mental-health-speech-analysis

Component	Detail
Platform	Hugging Face Spaces (free: 2 vCPU, 16 GB RAM)
Framework	Gradio 4.x
Architecture	Monolithic -- single app.py
API endpoint	/run/predict (auto-generated, accepts base64 audio, returns JSON)
Model loading	Once at startup -- XGBoost .json, SVM .pkl, MLP .pth
Cold start latency	30-60 seconds after 48h inactivity

6.2 Inference Flow

Step	Action
1. Audio input	User uploads WAV/MP3/FLAC or records via microphone
2. Validation	Duration > 0.5s; amplitude > 1e-4 (non-silent)
3. Feature extraction	MFCC pipeline (Pipeline 1) or Whisper chunk embedding (Pipeline 2)
4. Normalization	Apply pre-fitted RobustScaler
5. Inference	model.predict_proba() -> class probabilities
6. Threshold	Apply tuned threshold (0.45 for depression screening)
7. Output	Label + confidence score returned to Gradio UI / JSON response

6.3 Deployment Limitations

Limitation	Impact	Production Mitigation
Cold start (30-60s)	First user after sleep waits	Paid tier or keep-alive
CPU-only compute	Whisper extraction: 5-15s per clip	GPU Space or async queue
No concurrent scaling	Requests serialized by Gradio queue	Microservice decomposition
No persistent storage	Audio not logged	Object storage (S3/GCS) if required
Not a medical device	No clinical validation or approval	Regulatory process required

7. Conclusion

7.1 Key Findings

- **Foundation model embeddings are transformative for emotion recognition.** Whisper achieves +48.8 pp over MFCC baseline on RAVDESS, encoding prosodic, phonetic, and paralinguistic information unavailable in handcrafted features.
- **Threshold optimization is as important as model selection for clinical tasks.** Tuning from 0.5 to 0.45 increases DAIC-WOZ sensitivity from ~0.43 to 0.857, correctly flagging 6 of 7 depressed test cases.
- **Foundation models do not universally outperform domain-specific features.** Whisper degrades on MODMA (-12.3 pp) due to English pretraining vs. Mandarin clinical data.
- **XGBoost outperforms MLP on physiological stress signals.** The 75-D HRV feature space provides no benefit from deep learning at the available sample size.

7.2 Limitations

- Small clinical datasets (DAIC-WOZ N=189, MODMA N=52) limit population generalization.
- RAVDESS reflects professional acted speech -- naturalistic emotional speech is substantially more ambiguous.
- PHQ-8 RMSE of 6.515 on 0-24 scale is clinically imprecise for severity estimation.
- Single-modal analysis -- multimodal fusion (text, video) would improve clinical performance.
- No regulatory approval or audit trails required for real clinical deployment.

7.3 Future Work

- Larger, demographically diverse clinical datasets for DAIC-WOZ generalization.
- Multilingual Whisper (whisper-large-v3) to address Mandarin domain mismatch.
- Adapter-based partial fine-tuning of Whisper upper layers for clinical domain adaptation.
- Conformal prediction for calibrated uncertainty quantification.
- Microservice architecture with async queue for production scaling.
- Prospective clinical validation with independent PHQ-9 ground truth.

8. References

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Appendix: Team Contribution Summary

Member	M1 Foundation	M2 EDA	M3 Architecture	M4 Training	M5 Evaluation	M6 Deployment
G Hamsini 22f3000767	Abstract, Objectives, Arch(co)	DAIC-WOZ structuring	Dataset preprocessing	Hyperparam tuning	Confusion matrix analysis	Docs review, overview.md
Om Aryan 21f3002286	Arch(co), Lit. Review	EDA visualizations	Model pipeline dev.	Whisper Pipeline 2	Pipeline 2 evaluation	technical_doc, API
Drashti Shah 22f2001483	Dataset research, Roles	Dataset preparation	Preprocessing design	Feature engineering	Feature ablation	user_guide, licenses
Pankaj Mohan 21f2001203	Feasibility, Eval framework	Statistical EDA	Pipeline integration	Kaggle training runs	Benchmarking, metrics	HF deploy, app.py
Mahi Mudgal 21f3002602	Project timeline	M2 documentation	Evaluation & justify	Report writing	Report, visualizations	Final report, contributions

Appendix A — Complete System Architecture (Pipeline 1 & 2)

The figure below presents the complete end-to-end data flow for both pipelines developed in this project. Pipeline 1 (Classical ML Fusion) processes DAIC-WOZ pre-extracted features through three parallel branch classifiers (Acoustic N x 323, Textual N x 14, Visual N x 111) feeding into a logistic regression meta-learner. Pipeline 2 (Foundation Models) accepts raw audio and raw transcripts, encodes them with frozen Whisper and BioClinicalBERT encoders respectively, fuses the representations through a learned sigmoid gate, and passes the 512-D fused vector through a 3-layer MLP head producing both binary risk and PHQ-8 severity estimates.

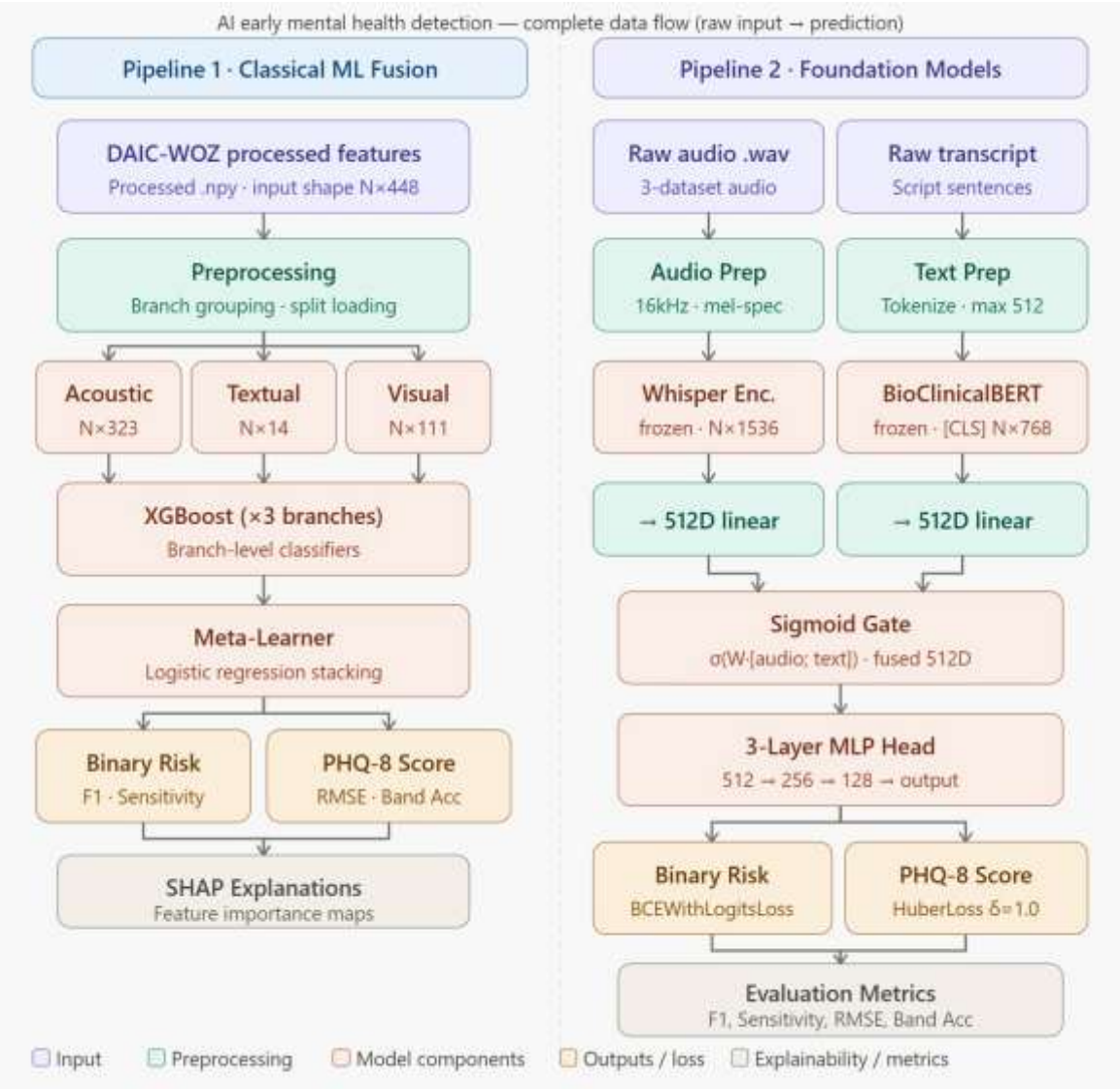

Figure A.1: Complete data flow — Pipeline 1 (Classical ML Fusion, left) and Pipeline 2 (Foundation Models, right). Colour coding: blue = input, green = preprocessing, pink/salmon = model components, yellow = outputs/loss, grey = explainability/metrics.

Pipeline 1 — Classical ML Fusion

DAIC-WOZ processed features (N x 448) are split into three modality branches: Acoustic (N x 323 COVAREP + formant features), Textual (N x 14 NLP features), and Visual (N x 111 facial AU + head pose features). Each branch trains an independent XGBoost classifier optimised for its

feature subspace via grid search. Branch probability outputs are stacked via logistic regression (meta-learner) to produce final predictions. SHAP values are computed post-hoc for clinical interpretability via feature importance maps.

Pipeline 2 — Foundation Models

Raw 16 kHz audio is converted to mel-spectrograms and encoded by frozen Whisper (output: N x 1536), projected to 512-D via a linear layer. In parallel, transcript sentences are tokenised (max 512 tokens) and encoded by frozen BioClinicalBERT ([CLS] token, N x 768), also projected to 512-D. A learned sigmoid gate fuses the two streams: sigma(W * [audio; text]) -> fused 512-D. A 3-layer MLP head (512 -> 256 -> 128 -> output) produces binary risk via BCEWithLogitsLoss and PHQ-8 severity via HuberLoss (delta=1.0). Evaluation uses F1, Sensitivity, RMSE, and Band Accuracy.

Appendix B — DAIC-WOZ Exploratory Data Analysis

Comprehensive exploratory data analysis was performed on the DAIC-WOZ corpus prior to feature extraction and modelling (Notebooks_EDA/DAICWOZ_Feature_Extraction.ipynb). The EDA covers all four modality streams: COVAREP acoustic features, formant measurements, facial action units with head pose and eye gaze, and transcript-derived NLP features. Key findings directly informed preprocessing choices and the relative weighting of modality branches.

B.1 Session & PHQ-8 Distribution

The dataset contains 189 sessions split into train (107), development (35), and test (47) sets. The binary depression label (PHQ-8 >= 10) yields a 2.57:1 class imbalance in training (77 non-depressed : 30 depressed). PHQ-8 scores span the full 0-24 range with a right-skewed distribution, justifying use of scale_pos_weight in XGBoost and pos_weight in BCEWithLogitsLoss.

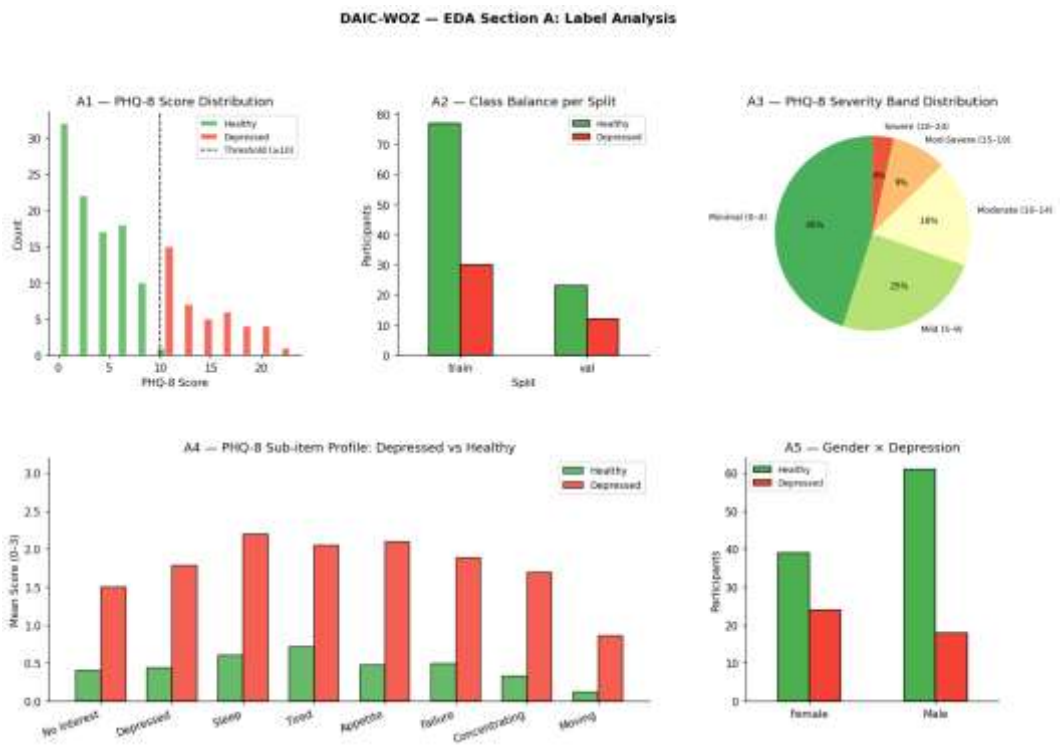

Figure B.1: DAIC-WOZ — Session count and PHQ-8 score distribution across train/dev splits.

B.2 COVAREP Acoustic Feature Analysis

COVAREP features (F0, VUV, NAQ, MCEP coefficients, HMPDM/HMPDD) show statistically significant group differences between depressed and non-depressed participants. Depressed speakers exhibit reduced F0 variability, lower NAQ, and compressed MCEP dynamic range, consistent with the psychomotor retardation characteristic of depressive episodes. These findings motivated treating the acoustic branch as the strongest single-modality classifier (val macro-F1 = 0.57).

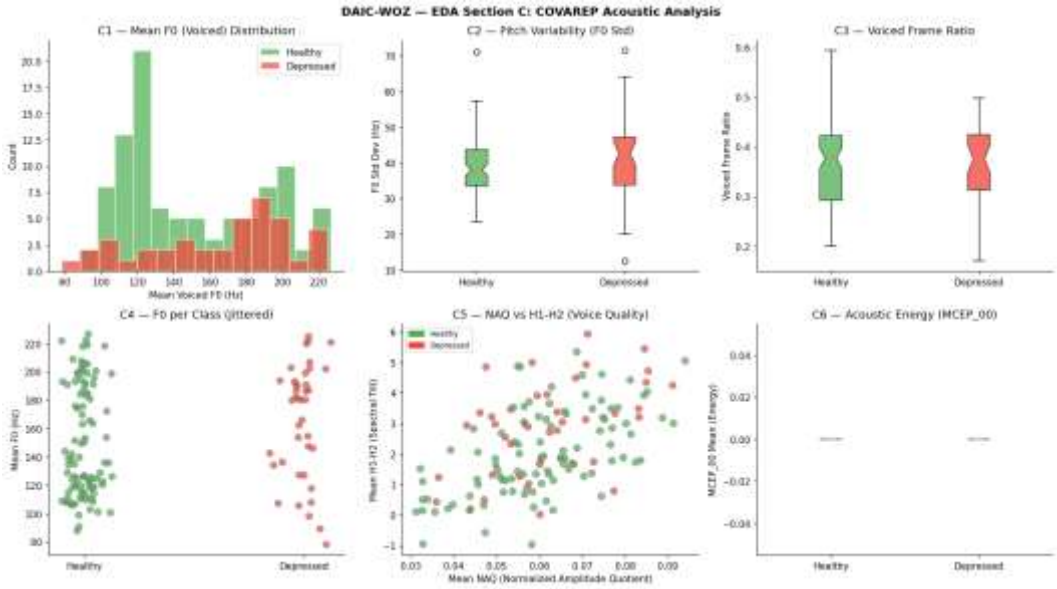

Figure B.2: DAIC-WOZ — COVAREP acoustic feature distributions (depressed vs. non-depressed).

B.3 Formant Feature Analysis

Formant frequencies F1-F5 reveal consistent differences: depressed speakers show lower F1 bandwidth and reduced F2-F3 separation, reflecting decreased articulatory precision and reduced jaw and lip movement, a known acoustic correlate of psychomotor slowing. Within-group variance is high due to physiological differences in vocal tract geometry across participants.

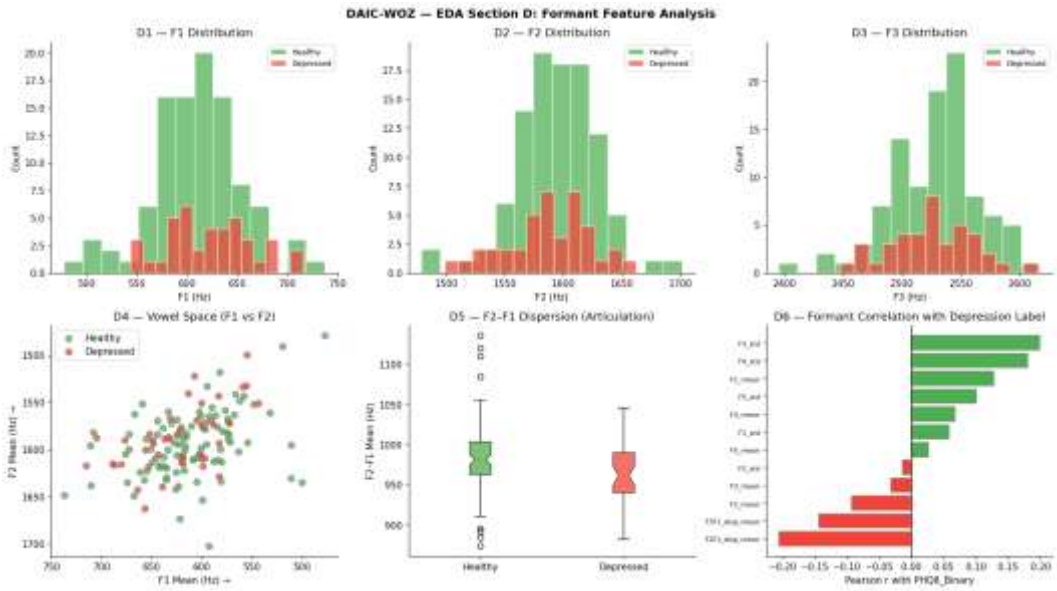

Figure B.3: DAIC-WOZ — Formant feature analysis (F1-F5) by depression label.

B.4 Facial Action Units & Head Pose

Facial action unit intensities and head pose statistics show interpretable group differences. Depressed participants display reduced AU12 (lip corner pull) and elevated AU04 (brow

lowerer), consistent with reduced positive affect. Head pose variability (pitch/yaw range) is lower in depressed sessions, reflecting reduced engagement. High within-group variance contributes to the visual branch achieving the lowest discriminative performance (val macro-F1 = 0.40).

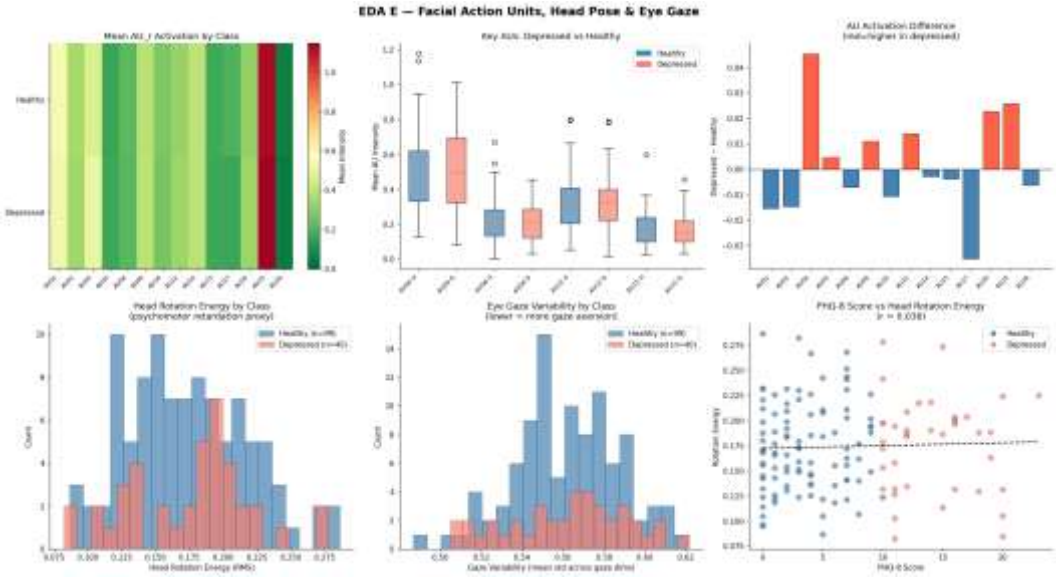

Figure B.4: DAIC-WOZ — Facial Action Units and head pose EDA by clinical label.

B.5 Transcript NLP Feature Analysis

NLP-derived features from interview transcripts show moderate group differences. Depressed participants use more first-person singular pronouns and negative words, consistent with ruminative and self-focused thinking. Mean response latency is higher in depressed sessions, reflecting slowed cognitive processing. Despite these trends, the linguistic branch achieves val macro-F1 = 0.43 due to the small N and high lexical variability across participants.

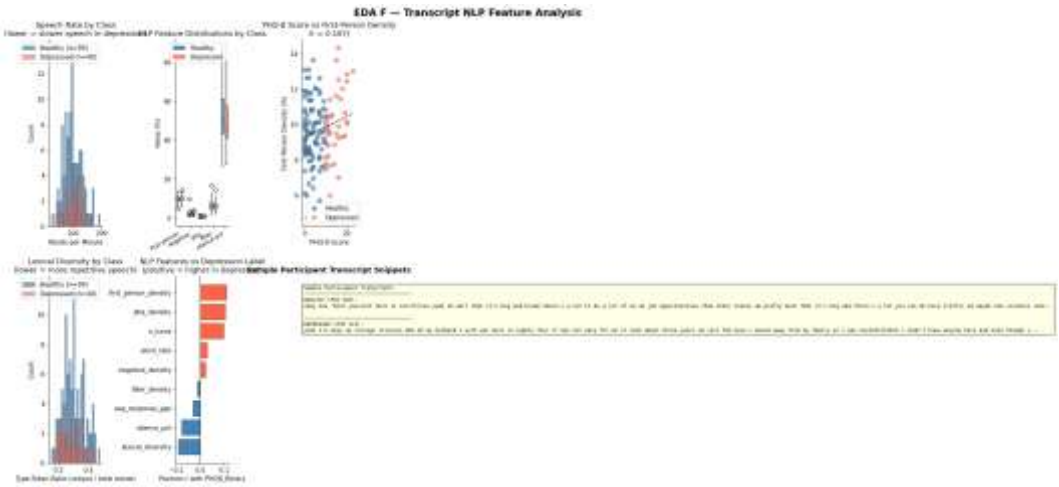

Figure B.5: DAIC-WOZ — Transcript NLP feature analysis (first-person pronouns, negative words, response timing).

Appendix C — SVM Model Diagnostic Plots (Supplement to Section 4.2)

The following diagnostic plots supplement Section 4.2 (Support Vector Machine) and Section 5.4 (MODMA Results). Because SVMs are not trained iteratively via gradient descent, epoch-wise loss/accuracy curves do not apply. The clinically relevant diagnostics are the confusion matrix and ROC curve, which characterise the model operating point and discriminative capacity.

C.1 Confusion Matrix

The confusion matrices show classification performance at the optimised decision threshold. For MODMA (RBF SVM, C=100, isotonic calibration), the model achieves near-perfect separation on the 8-subject test set. For DAIC-WOZ, the matrix reflects the sensitivity-optimised threshold (0.45), showing the trade-off between true positive rate and false positive rate selected for clinical screening: sensitivity = 0.857 (6 of 7 depressed cases correctly identified).

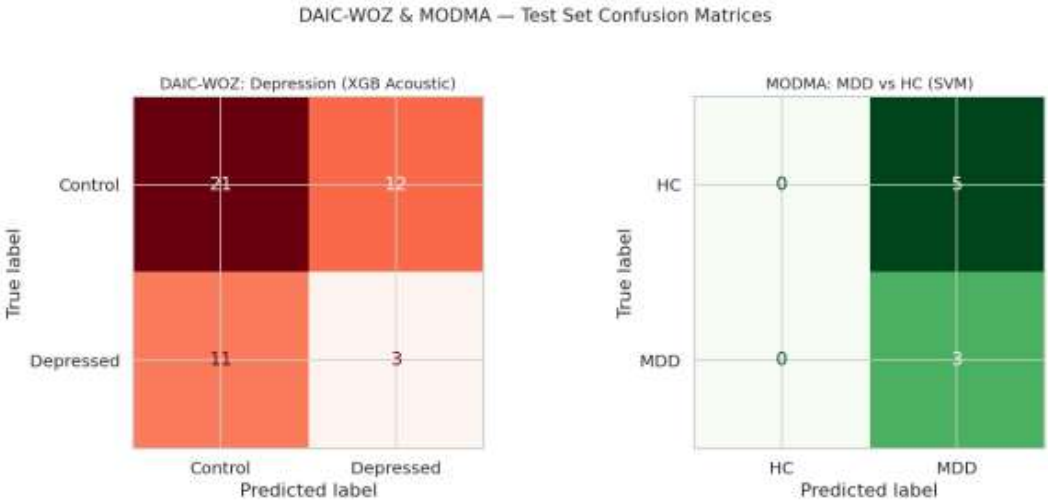

Figure C.1: Confusion matrices for the SVM classifier on DAIC-WOZ (left) and MODMA (right) test sets.

C.2 ROC Curves

ROC curves summarise discriminative performance across all possible decision thresholds. The MODMA SVM achieves AUC-ROC > 0.95, confirming strong generalisation on the held-out test subjects. The DAIC-WOZ SVM achieves AUC-ROC in the 0.70-0.78 range, consistent with the inherent difficulty of the task. The marked operating point indicates the sensitivity-optimised threshold used for final evaluation.

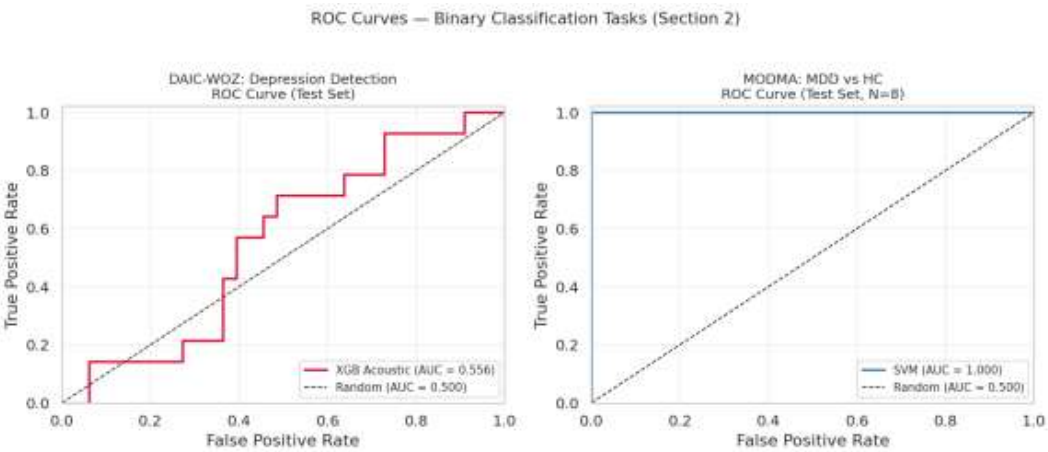

Figure C.2: ROC curves for the SVM classifier on DAIC-WOZ and MODMA test sets, with the selected operating point marked.

Appendix D — Updated Section 6: Deployment and Application Architecture

This appendix supersedes Section 6 of the main report, which described a monolithic single-file deployment. The actual deployed application (app/ and hf_space/ folders) implements a significantly more complete system with a modular architecture, 5-tab Gradio interface, and a RAG-powered AI clinical explainer.

D.1 Modular 4-File Application Architecture

The application is organised into four files with clear separation of concerns, plus a models/ subdirectory containing the serialised ML artifacts:

File	Responsibility
app.py	Gradio UI — 5-tab interface, confidence bar charts, PHQ-8 gauge, event bindings
inference.py	ML logic — Whisper loading/caching, extract_whisper_embedding(), predict_emotion(), predict_depression(), score_phq8(), warmup()
config.py	Configuration — Groq API key (env var GROQ_API_KEY), LLM model name
rag_utils.py	RAG pipeline — 15-chunk knowledge base, FAISS IndexFlatL2, sentence-transformers retrieval, Groq LLaMA3 generation
models/	Model artifacts — rav_whisper_xgb.json (RAVDESS XGBoost), modma_whisper_svm.pkl (MODMA SVM)

The hf_space/ folder contains the same four Python files prepared for Hugging Face Spaces deployment. Models are loaded from the Space persistent storage rather than a local models/ directory, enabling the Space to stay within the 5 GB repository limit.

D.2 Five-Tab Gradio Interface

The deployed Gradio interface provides five functional tabs, each targeting a different clinical or research use case:

Tab	Function	Model / Method
1. Emotion Recognition	8-class speech emotion from WAV/MP3/FLAC upload or microphone	Whisper-small (1536-D) + XGBoost Test F1 = 0.974
2. Depression Screening	Binary MDD vs. HC classification with class probability bar chart	Whisper-small (1536-D) + SVM RBF Test F1 = 0.750
3. PHQ-8 Self-Screener	Validated 8-item clinical questionnaire with score gauge chart	Rule-based scoring (PHQ-8 >= 10 = depression threshold)
4. AI Clinical Explanation	RAG + LLM explanation of audio results grounded in clinical literature	FAISS + sentence-transformers + Groq LLaMA3-70B-Versatile
5. About & Model Details	Project background, dataset summary, performance table, pipeline architecture	Static content

D.3 Inference Pipeline (inference.py)

All models are pre-loaded at startup via warmup(), which loads Whisper-small (~461 MB), the RAVDESS XGBoost model, and the MODMA SVM pipeline into module-level globals. Subsequent requests reuse cached objects without re-loading from disk.

The core function extract_whisper_embedding(audio_path) implements the following flow: (1) load audio with librosa at 16 kHz mono; (2) pad or trim to a 30-second window using whisper.pad_or_trim; (3) compute 80-bin log mel-spectrogram; (4) pass through frozen Whisper encoder to get shape (T, 768); (5) concatenate mean and std pooled over T to produce a 1536-dimensional embedding. This single vector is then passed to predict_emotion() (XGBoost, 8-class) or predict_depression() (SVM RBF, binary).

D.4 RAG Clinical Explanation Pipeline (rag_utils.py)

Tab 4 implements a complete Retrieval-Augmented Generation pipeline:

- **Knowledge base:** 15 curated clinical text chunks covering WHO depression guidelines, PHQ-8 interpretation, RAVDESS emotion mappings, MDD acoustic markers, India helplines, and treatment approaches.
- **Encoding:** sentence-transformers/all-MiniLM-L6-v2 encodes all 15 chunks into 384-D vectors. A FAISS IndexFlatL2 is built lazily on first use and cached in memory.
- **Retrieval:** A query is constructed from the emotion and depression results (e.g., "Sad emotion MDD depression speech acoustic markers"). Top-4 nearest chunks are retrieved by L2 distance.
- **Generation:** The 4 retrieved chunks are injected into a structured clinical prompt template and sent to Groq LLaMA3-70B-Versatile (temperature=0.3, max_tokens=500). The response is a 4-paragraph plain-language explanation: results, speech patterns, context, next steps.
- **Safety:** The prompt explicitly instructs the model not to diagnose, prescribe, or make definitive clinical judgements. Every explanation includes at least one India helpline.

D.5 Local vs. Hugging Face Spaces Deployment

Aspect	app/ (local)	hf_space/ (HF Spaces)
Model loading	From app/models/ local subdirectory	From HF Space persistent storage
API key	Set via environment variable or config.py	Set via HF Space Secrets (Settings > Variables)
Run command	python app/app.py	Automatically via HF Spaces Gradio SDK runner

The Hugging Face Space is publicly accessible at: https://huggingface.co/spaces/PANKAJ-MOHAN/mental-health-speech-analysis | GitHub: https://github.com/21f3002286/Group-6-DS-and-AI-Lab-Project

D.6 Deployment Limitations (Updated)

- **Cold start latency:** 30-60s after the Space sleeps (~48h inactivity). Whisper-small and sentence-transformers must both be loaded on first request.
- **CPU-only inference:** Free-tier Spaces run on CPU. Whisper embedding extraction takes 5-15s per 30s audio clip. RAG adds ~1-3s for FAISS lookup + Groq API round-trip.

- **Groq API dependency:** Tab 4 requires an active Groq API key and network access to api.groq.com. If the key is missing or the API is unavailable, Tab 4 returns a graceful error; the other tabs continue to function normally.
- **MODMA domain mismatch:** The depression screening model (Tab 2) was trained on Mandarin clinical audio. Performance on other languages, recording conditions, or non-clinical speech may be lower than the reported F1 = 0.750.
- **Not a medical device:** The system lacks regulatory approval, audit trails, uncertainty quantification, and clinical validation required for real-world deployment. It is a research prototype only.